

PRACTICAL 6

Title

Use String Functions and Regular Expressions for Email Validation, Data Extraction and Text Analysis

Aim

To implement JavaScript **string functions and regular expressions (Regex)** for email validation, data extraction, string manipulation, and text analysis.

Objectives

- To understand JavaScript string functions.
- To perform different string operations.
- To use Regular Expressions (Regex) for pattern matching.
- To validate email addresses.
- To extract numbers and words from text.
- To reverse a string.
- To select and count vowels from a string.
- To analyze text using JavaScript.

EXPERIMENT

Problem Statement

Create a web application using HTML, CSS, and JavaScript that performs the following operations:

1. Validate an email address using Regular Expressions.
2. Extract numbers from a given text.
3. Extract words from a given text.
4. Reverse the entered string.
5. Select vowels from the string.
6. Count the number of vowels.
7. Count the total number of characters.

Software Requirements

- Web Browser
- Visual Studio Code or any other code editor
- HTML
- CSS
- JavaScript

Files Required

Practical6/

```
|  
├── index.html  
├── style.css  
└── script.js
```

1. index.html

```
<!DOCTYPE html>  
<html lang="en">  
  
  <head>  
    <meta charset="UTF-8">  
  
    <meta name="viewport"  
      content="width=device-width, initial-scale=1.0">  
  
    <title>String Functions and Regex</title>  
  
    <link rel="stylesheet" href="style.css">  
  </head>  
  
  <body>  
  
    <div class="container">  
  
      <h1>String & Regex Analyzer</h1>  
  
      <!-- Email Validation -->  
  
      <h2>Email Validation</h2>  
  
      <label for="email">  
        Enter Email:  
      </label>  
  
      <input  
        type="text"  
        id="email"  
        placeholder="example@gmail.com"  
      >  
    </div>  
  </body>  
</html>
```

```
<button onclick="validateEmail()">
    Validate Email
</button>

<p id="emailResult"></p>

<hr>

<!-- Text Analysis -->

<h2>Text Analysis</h2>

<label for="textInput">
    Enter Text:
</label>

<textarea
    id="textInput"
    rows="5"
    placeholder="Enter your text here..."
></textarea>

<button onclick="analyzeText()">
    Analyze Text
</button>

<!-- Results -->

<div class="result">

    <p>
        <strong>Original Text:</strong>
        <span id="original"></span>
    </p>

    <p>
        <strong>Reversed Text:</strong>
        <span id="reversed"></span>
    </p>

    <p>
        <strong>Extracted Numbers:</strong>
        <span id="numbers"></span>
    </p>

    <p>
        <strong>Extracted Words:</strong>
        <span id="words"></span>
    </p>

    <p>
        <strong>Vowels:</strong>
        <span id="vowels"></span>
    </p>

    <p>
        <strong>Vowel Count:</strong>
    </p>

</div>
```

```
        <span id="vowelCount"></span>
    </p>

    <p>
        <strong>Total Characters:</strong>
        <span id="charCount"></span>
    </p>

</div>

</div>

<script src="script.js"></script>

</body>

</html>
```

2. style.css

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}

body {
    font-family: Arial, sans-serif;

    background-color: #f2f2f2;

    display: flex;
    justify-content: center;
    align-items: center;

    min-height: 100vh;
}

.container {
    width: 600px;

    background-color: white;

    padding: 30px;

    border-radius: 12px;

    box-shadow: 0 5px 15px rgba(0, 0, 0, 0.15);
}

h1 {
    text-align: center;

    color: #333;

    margin-bottom: 25px;
}
```

```
h2 {
  color: #007bff;

  margin-bottom: 15px;
}

label {
  display: block;

  margin-bottom: 8px;

  font-weight: bold;
}

input,
textarea {
  width: 100%;

  padding: 12px;

  margin-bottom: 15px;

  border: 1px solid #ccc;

  border-radius: 6px;

  font-size: 15px;
}

textarea {
  resize: vertical;
}

button {
  width: 100%;

  padding: 12px;

  border: none;

  border-radius: 6px;

  background-color: #007bff;

  color: white;

  font-size: 16px;

  cursor: pointer;

  margin-bottom: 15px;
}

button:hover {
  background-color: #0056b3;
}

hr {
  margin: 25px 0;

  border: none;
```

```

        border-top: 1px solid #ddd;
    }

    #emailResult {
        margin-top: 5px;

        font-weight: bold;
    }

    .result {
        margin-top: 20px;

        padding: 20px;

        background-color: #f5f8ff;

        border-radius: 8px;
    }

    .result p {
        margin-bottom: 12px;

        line-height: 1.5;

        word-wrap: break-word;
    }

    .result span {
        color: #333;
    }

```

3. script.js

```

// =====
// EMAIL VALIDATION
// =====

function validateEmail() {

    const email =
        document.getElementById("email").value;

    // Regular Expression for email validation

    const emailPattern =
        /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    const result =
        document.getElementById("emailResult");

    if (emailPattern.test(email)) {

        result.textContent =
            "Valid Email Address";

        result.style.color = "green";
    }
}

```

```

    } else {

        result.textContent =
            "Invalid Email Address";

        result.style.color = "red";
    }
}

// =====
// TEXT ANALYSIS
// =====

function analyzeText() {

    const text =
        document.getElementById("textInput").value;

    // Display original text

    document.getElementById("original").textContent =
        text;

    // =====
    // REVERSE STRING
    // =====

    const reversedText =
        text
            .split("")
            .reverse()
            .join("");

    // =====
    // EXTRACT NUMBERS
    // =====

    const numbers =
        text.match(/\d+/g);

    // =====
    // EXTRACT WORDS
    // =====

    const words =
        text.match(/[A-Za-z]+/g);

    // =====
    // SELECT VOWELS
    // =====

    const vowels =
        text.match(/[aeiouAEIOU]/g);

```

```

// =====
// COUNT VOWELS
// =====

const vowelCount =
    vowels ? vowels.length : 0;

// =====
// DISPLAY RESULTS
// =====

document.getElementById("reversed").textContent =
    reversedText;

document.getElementById("numbers").textContent =
    numbers
    ? numbers.join(", ")
    : "No numbers found";

document.getElementById("words").textContent =
    words
    ? words.join(", ")
    : "No words found";

document.getElementById("vowels").textContent =
    vowels
    ? vowels.join(", ")
    : "No vowels found";

document.getElementById("vowelCount").textContent =
    vowelCount;

document.getElementById("charCount").textContent =
    text.length;
}

```

Explanation of the Experiment

1. String Functions

JavaScript provides several built-in functions for manipulating strings.

split()

The `split()` function divides a string into an array.

```
text.split("");
```

For example:

Hello

becomes:

```
["H", "e", "l", "l", "o"]
```

reverse()

The `reverse()` function reverses the elements of an array.

```
text.split("").reverse();
```

join()

The `join()` function combines array elements into a string.

```
text.split("").reverse().join("");
```

Therefore, the complete statement reverses a string.

2. Regular Expressions

Regular Expressions are patterns used to search, match, and extract data from text.

Email Regex

```
/^[^\\s@]+@[^\\s@]+\\.[^\\s@]+$/
```

This pattern checks for a basic email structure containing:

```
username@domain.extension
```

The `test()` method checks whether the entered email matches the pattern.

```
emailPattern.test(email);
```

3. Extracting Numbers

Numbers are extracted using:

```
text.match(/\\d+/g);
```

Here:

- `\d` represents a digit.
- `+` means one or more occurrences.
- `g` searches throughout the complete string.

For example:

My age is 20 and my roll number is 105.

Output:

20, 105

4. Extracting Words

Words are extracted using:

```
text.match(/[A-Za-z]+/g);
```

This finds sequences containing English alphabetic characters.

5. Selecting Vowels

Vowels are selected using:

```
text.match(/[aeiouAEIOU]/g);
```

This matches:

```
a e i o u
A E I O U
```

6. Counting Vowels

The number of vowels is calculated using:

```
const vowelCount =
  vowels ? vowels.length : 0;
```

If vowels are found, `length` gives their total number.

If no vowels are found, the count is set to 0.

CASE STUDY

Case Study: String Operations

Aim

To develop a JavaScript program that performs basic string operations by reversing a string, selecting vowels, and counting the number of vowels.

Problem Statement

Consider a text string entered by the user.

The application should:

1. Accept a string.
 2. Display the original string.
 3. Reverse the string.
 4. Select all vowels.
 5. Count the vowels.
 6. Display the results.
-

Case Study Code

```
const text = "Hello World";

// Reverse the string
const reversed =
  text
    .split("")
    .reverse()
    .join("");

// Select vowels
const vowels =
  text.match(/[aeiouAEIOU]/g);

// Count vowels
const vowelCount =
  vowels ? vowels.length : 0;

// Display results
console.log("Original String:", text);
```

```
console.log("Reversed String:", reversed);  
console.log("Vowels:", vowels);  
console.log("Vowel Count:", vowelCount);
```

Case Study Input

Hello World

Case Study Output

Original String: Hello World

Reversed String: dlroW olleH

Vowels: e, o, o

Vowel Count: 3

Another Example

Input

JavaScript is Easy

Reversed String

ysaE si tpircSavaJ

Selected Vowels

a, a, i, i, E, a

Vowel Count

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Important JavaScript Functions Used

Function / Method	Purpose
<code>split()</code>	Converts a string into an array
<code>reverse()</code>	Reverses array elements
<code>join()</code>	Combines array elements into a string

Function / Method	Purpose
<code>match()</code>	Finds matching patterns using Regex
<code>test()</code>	Tests whether a string matches Regex
<code>length</code>	Finds the number of characters/elements
<code>textContent</code>	Displays content on the webpage

Regular Expressions Used

Regular Expression	Purpose
<code>/^[^\s@]+@[^\s@]+\.[^\s@]+\$/</code>	Email validation
<code>/\d+/g</code>	Extract numbers
<code>/[A-Za-z]+/g</code>	Extract words
<code>/[aeiouAEIOU]/g</code>	Select vowels

Expected Webpage Output

The webpage contains:

STRING & REGEX ANALYZER

Email Validation

Enter Email:
[example@gmail.com]

[Validate Email]

Valid Email Address

Text Analysis

Enter Text:
[Hello World 123]

[Analyze Text]

Original Text: Hello World 123
Reversed Text: 321 dlroW olleH
Extracted Numbers: 123
Extracted Words: Hello, World
Vowels: e, o, o
Vowel Count: 3
Total Characters: 15

Result

The experiment was successfully implemented using **HTML, CSS, and JavaScript**. String functions and regular expressions were used for email validation, data extraction, string reversal, vowel selection, vowel counting, and text analysis.

The case study successfully demonstrated practical string operations by reversing a string and selecting and counting its vowels.

Conclusion

This practical demonstrates the importance of **string manipulation and Regular Expressions in JavaScript**. String methods such as `split()`, `reverse()`, `join()`, and `length` can be used for text processing, while Regex provides an efficient way to validate and extract specific patterns such as email addresses, numbers, words, and vowels.